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Assignment – 7

Aim

To implement and analyze probability distributions—specifically Binomial, Poisson, and Normal distributions—using Python, and to understand their characteristics, differences, and real-world applications.

1. Difference between Binomial, Poisson, and Normal Distributions

Feature	Binomial Distribution	Poisson Distribution	Normal Distribution
Type	Discrete	Discrete	Continuous
Use Case	Fixed number of trials	Rare events over time/space	Natural continuous data
Parameters	n (trials), p (probability)	λ (average rate)	μ (mean), σ (std deviation)
Example	Tossing a coin	Number of calls per hour	Height of people
Shape	Can be symmetric/skewed	Right-skewed	Symmetric (bell-shaped)

2. Real-world significance of Poisson distribution

Examples:

Number of calls in a call center per minute
 Number of accidents on a road per day
 Number of typing errors per page 🐼 It helps in: Predicting probabilities of rare events
 Planning resources (like staffing in hospitals or call centers)
 Risk analysis in industries

3. Central Limit Theorem (CLT) and Normal Distribution

The Central Limit Theorem (CLT) states: $\bar{X} \sim N(\mu, \frac{\sigma}{\sqrt{n}})$

🐼 Meaning: When you take large samples from any distribution, The sample mean will be approximately normally distributed, Even if the original data is not normal.

Relation to Normal Distribution: CLT explains why Normal distribution appears everywhere It allows us to use Normal distribution for: Estimation Hypothesis testing

4. Why Normal distribution is widely used

The Normal distribution is popular because: It naturally occurs in many real-world datasets (height, weight, marks, errors, etc.) It is mathematically simple and well-understood Many statistical methods assume normality: Regression Hypothesis testing Machine learning algorithms Due to CLT, it works well even when original data is not normal

5. Skewness in distributions

Skewness means asymmetry in a distribution. Types: Positive skew (right-skewed) → tail on right side

Negative skew (left-skewed) → tail on left side
 Distribution | Skewness | | ----- | -----
 ----- | | Binomial | Can be skewed (depends on p) | | Poisson |

Positively skewed | | Normal | No skew (perfectly symmetric) | left side Zero skew → symmetric

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom, poisson, norm
```

```
n = 10
p = 0.5
```

```
x_binom = np.arange(0, n+1)
y_binom = binom.pmf(x_binom, n, p)
```

```
plt.figure()
plt.bar(x_binom, y_binom)
plt.title("Binomial Distribution")
plt.xlabel("Number of Successes")
plt.ylabel("Probability")
plt.show()
```

```
# ----- Poisson Distribution -----
lam = 3 # average rate
```

```
x_pois = np.arange(0, 15)
y_pois = poisson.pmf(x_pois, lam)
```

```
plt.figure()
plt.bar(x_pois, y_pois)
plt.title("Poisson Distribution")
plt.xlabel("Number of Events")
plt.ylabel("Probability")
plt.show()
```

```
# ----- Normal Distribution -----
mu = 0 # mean
sigma = 1 # standard deviation
```

```
x_norm = np.linspace(-4, 4, 100)
y_norm = norm.pdf(x_norm, mu, sigma)
```

```
plt.figure()
plt.plot(x_norm, y_norm)
plt.title("Normal Distribution")
plt.xlabel("Values")
```

```
plt.ylabel("Probability Density")  
plt.show()
```


Binomial Distribution

Conclusion:

0.25